

LexiSense: Formal Technical & Accuracy Verification Dossier

Unified Multimodal AI Framework Integrating Behavioral Triangulation, Speech Fluency, & Ocular Saccades

Executive Summary: LexiSense (v7.0) is an AI web platform engineered for early pre-diagnosis screening of developmental dyslexia in children aged 3–12. Synthesizing three screening pillars—a 30-item neurodevelopmental questionnaire (40%), an age-normed Oral Reading Fluency (ORF) WCPM Z-score engine (40%), and webcam ocular saccade tracking (20%)—the platform computes an objective Composite Dyslexia Risk Score (0–100%). Automated test suites empirically verify 100% mathematical accuracy across score calculations, edge-case N/A handling, and risk threshold categorization.

1. Scientific Screening Methodology & Mathematical Formulation

Composite Risk Score = (0.40 × Pillar 1) + (0.40 × Pillar 2) + (0.20 × Pillar 3)

- Pillar 1: Behavioral Questionnaire (40%)** — Evaluates 30 indicators across 5 domains. Uses dynamic denominator exclusion for N/A (-1) responses.
- Pillar 2: Oral Reading Fluency & WCPM Z-Score (40%)** — Quantifies WCPM against age norms (Year 4 Mean=105, SD=22) via $Z = (Expected - Observed) / SD$.
- Pillar 3: Ocular Saccades & Fixation (20%)** — Tracks webcam gaze using WebGazer.js to detect line regressions ($\geq 35px$) & fixation instability.

2. Clinical Risk Classification Matrix

Score Band	Risk Category	Clinical Interpretation	Recommended Pathway
0% – 34%	Few Indicators (Low)	Decoding & saccadic trajectories conform with grade milestones.	Routine classroom literacy; shared home reading.
35% – 64%	Some Indicators (Moderate)	Observable phoneme hesitations, line-skipping, or letter inversion.	Daily 10-min phonics; reading focus ruler; teacher check-in.
65% – 100%	Elevated Indicators (High)	Significant deficit convergence across questionnaire, WCPM, & regressions.	Formal referral to Educational Psychologist / Pediatrician.

3. Automated Verification Test Suite Results

Test Vector	Suite & Module	Verification Target	Status
Test 1: Low Risk Profile	Frontend (JS)	Assert 0% questionnaire score & 'Few Indicators' classification	PASS
Test 2: High Risk Profile	Frontend (JS)	Assert 100% questionnaire score & 'Elevated Indicators' classification	PASS
Test 3: N/A Exclude Math	Frontend (JS)	Assert 50% N/A answers exclude from denominator without distortion	PASS
Test 4: Multimodal Weighting	Frontend (JS)	Assert 3-pillar weighting (40% Q + 40% WCPM + 20% Gaze)	PASS
Test 5: Boundary Checks	Frontend (JS)	Assert strict 35% & 65% risk categorization thresholds	PASS
Backend: Django Model CRUD	Backend (Python)	Assert Child, User Roles, & AssessmentSession DB relations	PASS
Backend: Ollie AI Service	Backend (Python)	Assert context-aware prompt building & local knowledge engine	PASS

4. Architecture & Security Compliance

- Ephemeral Video Processing:** Webcam frames processed in client RAM via WebGazer.js; zero raw video stored (COPPA/FERPA compliant).
- Multi-Tenant Data Security:** Supabase PostgreSQL Row-Level Security (RLS) policies & Django DRF SimpleJWT authentication.
- Accessibility Standards:** Complies with WCAG 2.1 AA via OpenDyslexic fonts, focus ruler, and typography scaling.